

What changes when the facility changes

Diagram 03 • Failure response

For each condition, the map shows ownership, operator context, DUNE response, and validation evidence.

FACILITY CONDITION		WHAT OPERATORS SEE	WHAT DUNE DOES	VALIDATION EVIDENCE
 Power and reserve ELECTRICAL • SURF	Utility, transfer switch, generator, backup-supply state, and remaining reserve	An orderly data-acquisition stop is requested. This operational path is not a detector trip.	EVIDENCE Transfer test Loaded UPS runtime	
 Network or BMS link NETWORKING • CONTROLS	The failed link and affected points; values change to UNKNOWN or STALE	Detector and local protection remain available. Stale facility data is never treated as current.	EVIDENCE Per-link failure injection Recovery-behavior verification	
 Fire, low oxygen, or ventilation FACILITY SAFETY • SURF	Source alarm, trouble, bypass, area, acknowledgement, and time	Facility teams handle people and facility response. DPS applies only an approved mapped action.	EVIDENCE Zone-by-zone test Trouble and bypass	
 Cryogenic or liquid-argon state CRYOGENICS • DETECTOR SAFETY	Named level, pressure, relief, boiling-risk, and oxygen-deficiency states	The response follows the approved HV or DPS mapping. No generic liquid-argon leak signal is assumed.	EVIDENCE Source definition Cause-and-effect test	
 Cooling or water FACILITIES • EQUIPMENT OWNER	Area, affected equipment, water or cooling state, and owner	Rack or equipment-local protection acts first. DPS acts only where the matrix requires it.	EVIDENCE AHU, water, and sump Local-sensor test	
 Weather or access restriction SURF OPERATIONS	Defined access, shaft, power, cooling, or communications restriction	Raw weather provides context only. Action follows a defined SURF operational state.	EVIDENCE Tabletop scenario Operational-state injection	
Orderly stop • data acquisition and automation Uses tested UPS reserve for a controlled shutdown.		Detector trip • Detector Protection System Applies approved cause and effect; latches where required.	Local protection • rack or equipment Acts close to the threatened load and reports status upward.	